Fair Test Practical · Staff Answers

Learning goal: I can record repeated results, spot an odd reading, and use the data to answer our question.

Arrival Answers

1. Air resistance.

2. Yes. They are very close.

3. An anomaly.

4. Yes. You can ask or pass.

Starter & Hinge Answers

Starter: A. A larger canopy area increases air resistance opposing the fall, so it falls more slowly.

Hinge: A. 1.3 s is the anomalous reading caused by the unopened canopy; the valid trials cluster around 3.4 s.

Evidence Desk Cards

Small canopy: 1.2 s, 1.3 s, 1.1 s (typical 1.2 s): Consistent repeat data: all three readings are within 0.2 s.

Large canopy: 3.4 s, 3.5 s, 3.3 s (typical 3.4 s): Consistent repeat data: gives a clear typical fall time of 3.4 s.

Trial where strings twisted: 1.3 s instead of 3.4 s: Anomalous reading: circle it in the table and repeat the drop with untangled strings.

30x30 cm took 3.4 s vs 1.2 s for 10x10 cm due to air

resistance: Strong conclusion: quotes the times and explains that larger area increases air resistance.

Stopwatch pressed slightly late by hand: Method evaluation: use a clear '3-2-1-drop' signal or slow-motion camera.

Draught from an open window blew the canopy sideways: Method evaluation: close windows so air currents do not disturb the drop.

Check & Repair Answer

Scientists keep all raw data honest; circling the anomaly and recording a repeat shows good scientific method.

Exit Ticket Answers

1. 30x30 cm (3.4 s). 2. Its larger surface area meets more air particles, producing greater upward air resistance. 3. Keep the reading visible, circle/flag it as an anomaly, and carry out a clean repeat drop.